PingSheng Kao

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SUMMARY

Machine Learning Engineer and ML Data Scientist with 6+ years of experience building automated evaluation and monitoring for multimodal, LLM, agentic, and production ML systems. Specializes in scalable judges, auditable measurement, fairness analysis, failure discovery, and decision-ready reporting that guides model mitigation.

EXPERIENCE

ML Data Scientist - Apple July 2020 - Present
Santa Clara, California

- Scaled LLM- and agent-as-judge evaluation on an existing internal batch platform, resolving throughput and rate-limit issues, shipping versioned evaluator updates, and surfacing results through dashboards.
- Built simulated-user, agent-as-judge, targeted-scenario, and behavioral-metric workflows for a multimodal conversational agent, evaluating intent alignment, inferred user state, factual consistency, and tool-call behavior.
- Developed a reference-based, multi-stage LLM evaluator used in an internal production workflow, combining expert-calibrated gold sets with human-in-the-loop validation; reached 90% expert alignment on 40 examples.
- Designed Bayesian per-subject fairness analysis using Beta posteriors to estimate demographic-gap risk against a 10% product threshold.
- Built SQL/AWS Athena metric pipelines and leadership-facing model-release reports combining failure-pattern analysis, regression attribution, and online validation of offline-mined risks.
- Personally fine-tuned vision models on VLM-discovered and relighting-derived failure slices, comparing targeted model repair with alternative slice-discovery approaches.

RESEARCH PUBLICATION

GH-ESD: Grounded Hypothesis-Driven Error Slice Discovery for Instance-Level Vision Tasks ECCV 2026
Listed author (personal contribution: experiments) - Apple ML Research

- Listed author and experiment contributor on the ECCV 2026 publication GH-ESD, developing grounded, hypothesis-driven error-slice discovery for instance-level vision tasks.

ADDITIONAL EXPERIENCE

Algorithm and Architecture Intern - Deep Learning Compression - Ambarella June 2019 - August 2019
Santa Clara, California

- Developed a saliency-based neural-network quantization heuristic that reduced an exponential search formulation to a linear-time method, and proposed a mixed-precision algorithm integrated into company software.

EDUCATION

M.S. in Computer Science and Engineering - University of Michigan, Ann Arbor 2018 - 2020
B.S. in Electronic Engineering - National Taiwan University 2013 - 2017
GPA: 3.9/4.0 (B.S.)

SKILLS

Programming & Data: Python, SQL, C++, JavaScript, AWS Athena, Tableau
AI Safety Evaluation: Agent-as-Judge, LLM-as-Judge, Simulated Users, Tool-Call Evaluation, DSPy
Monitoring & Assurance: Evaluator Versioning, Dashboards, Gold Sets, Human-in-the-Loop Validation
Methods: Bayesian Fairness, Experiment Design, Embeddings, Clustering, OOD Analysis, Model Fine-Tuning